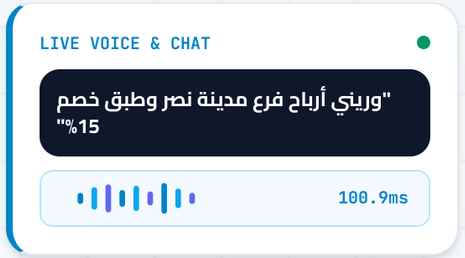
HITU — FACULTY OF AI & DATA SCIENCE — CLASS OF 2026

Multi-Tenant SaaS | LangGraph + LightGBM | Egyptian Dialect NLP

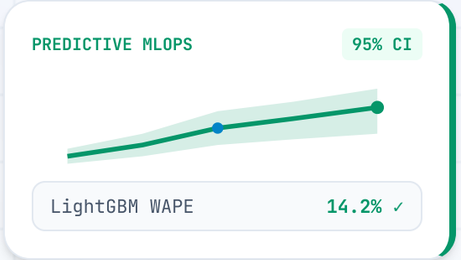


TalkShop

GRADUATION PROJECT DEFENSE DECK • FACULTY OF AI 2026

Next-Gen Multi-Tenant Agentic E-Commerce & ERP OS

Bridging Egyptian Colloquial Arabic, Autonomous Relational Text-to-SQL & Predictive MLOps for Retail



GRADUATION PROJECT ENGINEERING TEAM

فريق عمل مشروع التخرج • دفعة 2026

01 **Moustafa Maher** مصطفى ماهر

02 **Adel Ahmed** عادل أحمد

03 **Abdallah Sameh** عبد الله سامح

04 **Retaj Hakim** ريتاج حكيم

05 **Renad Hakim** رناد حكيم

06 **Roaa Ashraf** رؤى أشرف

SLIDE 02 • TALKSHOP EXECUTIVE SUMMARY & THESIS

The Paradigm Shift: From Traditional Software to Agentic Execution

Static CRUD Forms → TalkShop Command Center

LEGACY ERP (12+ SCREENS)
Manual Forms, SQL & Static Grids
4.5 hrs/day wasted on context switching

TALKSHOP AGENTIC BRIDGE

Dialect NLP + LangGraph + LightGBM

TALKSHOP OS (1 INTERFACE)
Autonomous Execution + GenUI Cards
Sub-120ms Verified Action Execution

Pillar 01 • NLP

Native Dialectal Grounding

Zero-API understanding of Egyptian colloquial Arabic and Franco-Arab business idioms with deterministic schema mapping and zero numeric hallucination.

SUPPORTED EGYPTIAN LEXICON

"عايز أعرف" "وريني السيلز" "رصيد الستوك" "فرع مدينة نصر"

Pillar 02 • Safety

Autonomous Execution with HITL

Safe database mutations (Bulk discounting, Store provisioning, Targeted promo release) guarded by a mandatory **Human-in-the-Loop** margin protection gate.

MARGIN FLOOR INVARIANT

`New_Price > Cost_Price` ✓ Enforced

Pillar 03 • MLOps

Integrated Predictive MLOps

Seamless union of live transactional OLTP records with quantile LightGBM demand forecasting (95% CI) and Log-RFM K-Means customer intelligence.

PRODUCTION ML MODELS

LightGBM ($\alpha \in \{.025, .5, .975\}$) — WAPE 14.2%

KEY BENCHMARK **Over 71 relational tables in AdventureWorks & Olist Egypt schemas unified in one conversational interface.**

91 Foreign Keys • Zero Manual SQL

SLIDE 03 • MARKET PROBLEM STATEMENT

Operational Friction in Traditional ERP & E-Commerce

⚠ Egypt & MENA Retail Pain Points

FRICTION 01 • CONTEXT SWITCHING

Administrative Fragmentation

4.5 hrs/day lost

Merchants waste **4.5 hours daily** switching between **12+ disconnected dashboards** (Inventory, POS, Shopify, Excel sheets, Bosta shipping, and Paymob portals).

POS ↔ Excel ↔ Bosta ↔ Paymob ↔ 56% Day Wasted

Impact: Delayed pricing & fulfillment errors

تشتت بين 12+ لوحة تحكم منفصلة

FRICTION 02 • DATA INACCESSIBILITY

Relational BI Blindspot

71 Tables / 91 FKs

Extracting net profit drivers requires complex 5-table SQL joins separating unit price, discount, and cost. Non-technical store owners operate blind without real-time BI.

```
SELECT SUM((oi.unit_price - p.cost) * oi.qty) FROM orders o JOIN order_items oi...
```

Impact: Gut-feeling decisions instead of data

استحالة كتابة استعلامات SQL للتاجر

FRICTION 03 • SUPPLY CHAIN

Stockout & Inventory Crisis

-23% Revenue

23% of Egyptian retail revenue is lost due to reactive static inventory reordering and unpredicted seasonal demand spikes (Ramadan, Eid, and White Friday).



Impact: 23% Lost sales & dead capital

خسارة 23% بسبب نفاد المخزون المفاجئ

FRICTION 04 • LOCALIZATION FAILURE

The NLP Dialect & Legal Gap

Law 181/2018

Off-the-shelf global LLMs fail to map local Egyptian trade idioms, Franco-Arab slang, and mandatory Egyptian Consumer Protection regulations (Law 181/2018).

✓ TalkShop: مطابق للمادة 17 و21

✗ ChatGPT: لا يفهم "الأوردرات المرتجعة"

Impact: High error rates on Arabic prompts

فجوة فهم العامية المصرية والقانون المحلي

TalkShop

SLIDE 04 • ARCHITECTURAL DISTINCTION

Traditional Chatbots vs. Autonomous AI Agents

Function Calling + State Graphs + Guardrails

FEATURE DIMENSION	TRADITIONAL CHATBOTS (THE FAILURE MODE)	TALKSHOP AGENTIC SYSTEM (OUR SOLUTION)
Core Role	Static Text Generation (FAQ): Passively answers questions from fixed text without operational agency.	Execution & Action Privileges: Builds stores, mutates prices, triggers ML models, and issues promos.
Data Access	No DB / Unsafe Raw SQL: Disconnected from live ERP state or vulnerable to destructive SQL injection.	Strict AST-Validated SQL Execution: 5-layer `sqlglot` sandboxed read-only analytical execution.
Safety & Writes	Hallucinates Numbers & Prices: Invents discounts that violate cost floors and destroy store profitability.	Human-in-the-Loop Margin Protection: Dry-run simulation enforcing $\text{New Price} > \text{Cost Price}$ before commit.
UI Experience	Isolated Plain-Text Chat Bubble: Dumps hard-to-read markdown walls separated from the dashboard.	Generative UI (GenUI): Streams interactive React cards (charts, diff tables, approval buttons) directly in chat.
Multi-Tenancy	Vulnerable to Context Leaks: Prompt injection can trick the bot into exposing competing merchants' data.	PostgreSQL Kernel RLS: Cryptographic tenant isolation via `app.current_tenant_id`.

"A Chatbot talks; an AI Agent operates, verifies, and executes."

الشات بوت يتحدث فقط.. الوكيل الذكي يخطط، يتحقق، وينفذ بأمان

TalkShop

SLIDE 05 • CORE RESEARCH & SYSTEMS CHALLENGES

Overcoming the 4 Critical Engineering Hurdles

NLP • State Machines • MLOps • Security

01

SCHEMA COMPLEXITY

Text-to-SQL on 71 Tables & 91 Foreign Keys

Dumping 71 ERP tables into an LLM prompt causes context overflow and hallucinated JOIN paths.

OUR ENGINEERING SOLUTION

Dynamic Vector Schema Scoping prunes 71 tables into **2–4 target relations** while preserving full FK JOIN fidelity.

02

AGENT DETERMINISM

Breaking Unbounded ReAct Loops

Standard autonomous agents get stuck in infinite tool-calling loops or execute unapproved state mutations.

OUR ENGINEERING SOLUTION

Deterministic Cyclic State Graphs (**LangGraph**) with bounded retries (max 3) and persistent `MemorySaver` checkpoints.

03

STATISTICAL MLOPS

Intermittent & Asymmetric Demand Forecasting

Retail SKU sales exhibit zero-inflated intermittent demand and non-Gaussian variance during local holidays.

OUR ENGINEERING SOLUTION

Non-parametric **LightGBM Quantile Regression** ($\alpha \in \{0.025, 0.5, 0.975\}$) with calendar holiday weights & EWMA velocity.

04

MULTI-TENANT SECURITY

Cryptographic Multi-Tenant Isolation

Competing merchants share database infrastructure; a single missing `WHERE` clause could leak rival financial data.

OUR ENGINEERING SOLUTION

PostgreSQL 16 Kernel **Row-Level Security (RLS)** bound to verified JWT `tenant_id` session contexts.

SLIDE 06 • MASTER ARCHITECTURAL BLUEPRINT

TalkShop End-to-End 5-Layer System Architecture

⚡ Sub-120ms p95 Latency • OLTP / OLAP Separation

LAYER 01 • PRESENTATION **Client & Generative UI**	React 18 + Vite + Tailwind	GenUI Component Registry	Redux Toolkit + React Query	Recharts + RTL i18next
LAYER 02 • API & SECURITY **Gateway & Firewall**	FastAPI + NestJS Gateway	PromptGuard & PII Masking	JWT TenantContextGuard	SSE Streaming (`X-Trace-ID`)
LAYER 03 • AGENTIC CORE **LangGraph Orchestrator**	MasterCommandGraph (10 Routes)	3-Tier Memory (Session/Pref/Audit)	HITL Interrupt & Margin Gate	Langfuse Distributed Tracing
LAYER 04 • MLOPS ENGINE **Predictive Microservices**	LightGBM Quantile Regressor	SHAP TreeExplainer Attributions	Log-RFM K-Means ($K = 5$)	Celery Workers + MLflow S3
LAYER 05 • PERSISTENCE **Data & Search Kernel**	PostgreSQL 16 + Kernel RLS	Redis 7 Cache & Idempotency	Meilisearch v1.6 Bilingual	SQLite Zero-Config Dev Mode

Monorepo Workspaces: apps/web • apps/api • apps/agent • apps/ml • backend • packages/database

Complete Separation of OLTP Transactions from Analytical Feature Views (v_timeseries_daily_sales)

Deterministic Orchestration & The 4-Step HITL Protocol

Lead: Moustafa Maher • MasterCommandGraph

LANGGRAPH STATE MACHINE

MasterCommandGraph Flow

1 **Guardrail & PromptGuard Node**
Regex & token screening for adversarial injection / PII

↓

2 **3-Tier Context Injector Node**
Recalls Session history, Merchant Preferences & Tenant context

↓

3 **Semantic Intent Router [10 Routes]**
Routes to SQL BI, Forecast, Catalog, RFM, Promo, Policy RAG

↓

4 **Specialized Tools → HITL Interrupt Gate**
Pauses state-mutating operations until merchant approval

Checkpointer: MemorySaver

Zero Unbounded Loops

01 Dry-Run Mutation & Margin Floor Check
Computes price delta across target SKUs without writing to DB; verifies gross margin rule.

New Price > Cost Price

02 State Checkpoint & Execution Interrupt
Saves proposed payload in PostgreSQL `agent_actions` table and halts graph thread.

PENDING_CONFIRMATION

03 Generative UI Interactive Preview Render
Frontend receives typed SSE event and renders interactive React diff card with margin badges.

<BulkPricePreviewCard />

04 Cryptographic Confirmation & Atomic Commit
On merchant click ("Approve"), backend verifies token, commits SQL transaction, and invalidates Redis.

✓ Redis Idempotency Lock

SLIDE 08 • CONVERSATIONAL BUSINESS INTELLIGENCE

Secure 5-Layer AST-Sandboxed Text-to-SQL Pipeline

sqlglot AST Verification • Read-Only Sandbox

Layer 01

Dialect Normalizer

Strips tashkeel, unifies Alef/Ya variants, and maps Egyptian slang into canonical business entities.

"وريني أرباح فرع مدينة نصر"

Layer 02

Dynamic Schema Scoping

Vector-prunes 71 ERP tables down to the exact target relations needed for the query.

```
SalesOrderHeader, Product,
Customer
```

Layer 03

SQL AST Parsing

`sqlglot` parses AST and verifies root node is strictly `exp.Select`. Blocks DDL/DML & catalogs.

🚫 Rejects DROP, UPDATE, DELETE, sys

Layer 04

Row-Clamp Guardrail

Enforces mandatory row limits and injects cryptographic tenant isolation filter into AST.

```
LIMIT 100 + tenant_id = :id
```

Layer 05

Self-Correction Loop

Catches database execution tracebacks and automatically refactors JOIN/column syntax (up to 3 retries).

98.5% Valid SQL Rate (VSR)

ADVERSARIAL RED-TEAM EVALUATION

100.0% Security Compliance (20/20 Destructive Adversarial Attacks Blocked)

SQL Injection • Piggybacked Semicolons • Schema Probing
All Intercepted Pre-Database Pool

Machine Learning in Retail: Demand Forecasting & RFM

Leads: Retaj Hakim (Forecasting) & Renad Hakim (RFM)

Panel A • Demand Forecasting (Retaj Hakim)

WAPE = 14.2%

LightGBM Quantile Regression & SHAP

Multi-quantile gradient boosting ($\alpha = 0.025, 0.5, 0.975$) generating median demand forecasts bounded by a 95% Confidence Interval.

SHAP TREEEXPLAINER FORECAST DECOMPOSITION

Price Discount Feature	+65 units
Ramadan / Eid Seasonality	+42 units
Day-of-Week Effect	-12 units

SAFETY STOCK & REORDER POINT

$$SS = Z_\alpha \times \sigma_L, \quad ROP = \mu_L + SS$$

EMPIRICAL ACCURACY

WAPE: 14.2% | Coverage: 94.8%

Panel B • Customer Intelligence (Renad Hakim)

Silhouette = 0.5828

Log-Transformed RFM & K-Means ($K = 5$)

Corrects right-skewed monetary distributions via $\text{StandardScaler}(\log(1 + [R, F, M]))$ clustered into 5 actionable behavioral personas:

Champions	Loyal	Potential	At-Risk	Hibernating
High R, F, M	Consistent	Recent Buyers	Churn Alert	Inactive

CLUSTER SEPARATION QUALITY

Silhouette Score: 0.5828 (Surpasses 0.40 Academic Benchmark)

+45.7% Above Target

🎯 Automated Win-Back Coupon Trigger (At-Risk Segment) — HITL Approval Gate

TalkShop

SLIDE 10 • GENERATIVE UI (GENUI) COMPONENT REGISTRY

Transforming Chat into Dynamic Interactive React Widgets

Lead: Abdallah Sameh • Zero Plain-Markdown Blobs

<BulkPricePreviewCard />

Category: Speakers (-15%)

Margin Safe ✓

SKU / Item	Old Price	New Price	Gross Margin
JBL Flip 6	~~4,800 EGP~~	4,080 EGP	28.9%
Anker Soundcore	~~2,200 EGP~~	1,870 EGP	31.4%

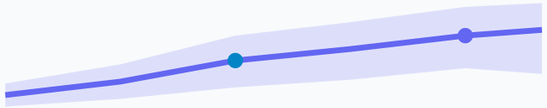
14 SKUs Affected • Cost Floor Verified

Cancel ✓Commit Changes

<SalesForecastCard />

Horizon: 30 Days

Restock Alert: 9 Days



Predicted Demand

+342 Units

95% CI: [298-389]

SHAP Top Driver: Discount (+65)

ROP = 145 Units

<CustomerSegmentsCard />

K-Means (K=5)

Silhouette: 0.5828



- Champions (28%)
- Loyal (24%)
- At-Risk (19%)
- Potential (29%)

412 At-Risk Customers Identified

🚀 Launch Targeted Promo

<InventoryAlertsList />

Stockout Velocity Engine

3 Critical SKUs

Anker 65W GaN Fast Charger
Stock: 12 units | ROP: 45 units
3.2 Days to Zero

Samsung T7 Shield 1TB SSD
Stock: 19 units | ROP: 35 units
5.8 Days to Zero

Courier: Bosta Express Restock

📦 Issue Supplier PO

SLIDE 11 • LIVE PRODUCTION WORKFLOWS

Dialectal NLP & Autonomous Operations in Action

من الأمر بالعامية المصرية إلى التنفيذ الفوري

Scenario 01 • Operational

MERCHANT PROMPT

"طبق خصم 15% على السماعات اللي رصيدها أكتر من 50"

01. Extracts intent `bulk_update_prices`, category=`Speakers`, stock > 50.
02. Executes dry-run verifying **New Price > Cost Price** across 14 matching SKUs.
03. Renders `<BulkPricePreviewCard />` and updates 14 SKUs atomically upon approval.

Execution Mode — HITL Guarded Write

Scenario 02 • Analytical BI

MERCHANT PROMPT

"عايز أعرف مبيعات فرع التجمع مقارنة بفرع المعادي الشهر اللي فات"

01. Normalizes Egyptian location entities (New Cairo / Maadi) & relative date window.
02. Generates AST-validated 3-table JOIN SQL query with `tenant_id` filter.
03. Formats currency in EGP and streams interactive Recharts Bar Chart + Arabic insight.

Execution Mode — AST Sandboxed SELECT

Scenario 03 • Legal & Policy RAG

MERCHANT PROMPT

"العميل عايز يرجع سماعة بعد 20 يوم.. هل ده قانوني؟"

01. Queries Store Policy RAG indexed with **Egyptian Consumer Protection Law 181/2018**.
02. Distinguishes **Article 17** (14-day return without defect) vs. **Article 21** (30-day return if defective).
03. Advises merchant: Only eligible if manufacturing defect exists; cites official legal clause.

Execution Mode — Citation-Grounded RAG

TalkShop

SLIDE 12 • ECOSYSTEM LOCALIZATION

Tailored for Egyptian Market Realities

Leads: Adel Ahmed (Integrations) & Roaa Ashraf (Policy RAG)

FinTech Gateway

Payment Infrastructure (Paymob)

Native EGP checkout flow supporting Egypt's dominant digital and cash-reference payment rails with webhook verification.

- Meeza National Cards
- InstaPay Transfers
- Fawry Reference Code
- Visa / Mastercard

✓ HMAC Webhook Verification & Auto Order State Sync

Last-Mile Logistics

Smart Logistics (Bosta Integration)

Automatic Air Waybill (`shipping_awb`) generation upon order confirmation with live tracking and governorate-based SLAs:

Cairo & Giza Hub	24 Hours ETA
Alexandria & Delta	48 Hours ETA
Upper Egypt & Red Sea	72 Hours ETA

✓ Real-time <OrderTrackingCard /> in Chat

Regulatory Compliance

Legal Guardrails (Law 181/2018)

Policy RAG engine pre-loaded with official Egyptian Consumer Protection statutes protecting both merchant and consumer rights.

14-Day Standard Return Window
Unopened merchandise in original packaging with invoice.

30-Day Defective Item Protection
Full refund or replacement for verified manufacturing defects.

✓ Zero-Hallucination Legal Article Citations

TalkShop

SLIDE 13 • AIR-GAPPED EDGE AI DEPLOYMENT

100% Offline Zero-API Architecture (Local GGUF Runtime)

Zero Cloud Fees • 100% On-Premise Data Sovereignty

QLoRA Fine-Tuning + GGUF

Custom Fine-Tuned Egyptian SQL Model

Fine-tuned specifically on Egyptian colloquial Arabic and Franco-Arab Text-to-SQL schemas, eliminating dependence on external OpenAI or Google cloud APIs.

BASE FOUNDATION MODEL	Qwen2.5-Coder-3B-Instruct
ADAPTATION METHOD	QLoRA (4-bit NormalFloat)
QUANTIZED ARTIFACT	qwen2.5-coder-3b-egyptian.Q4_K_M.gguf

Runtime Engine: llama.cpp / llama-cpp-python

Air-Gapped Ready

Edge Hardware Benchmark

Consumer Hardware Feasibility

Engineered to run smoothly on standard merchant laptops and affordable local office workstations without costly enterprise A100 GPUs.

VRAM FOOTPRINT	INFERENCE SPEED	OPERATING COST
2.2 GB	38 tok/s	$0.00
Fits on 4GB GPU	CUDA Accelerated	Zero Token Fees

Verified Target GPU: NVIDIA GeForce RTX 3050 (4GB VRAM) ✓ 100% Local

Zero External Network Calls Required

Enterprise Privacy Compliant

SLIDE 14 • EMPIRICAL EVALUATION & QUALITY ASSURANCE

Academic Benchmarks & Automated Slice Verification

100-Prompt Evaluation Dataset • 6 Unittest Suites

SECURITY COMPLIANCE

100.0%

20/20 Destructive Adversarial SQL & Prompt Injection Attacks Blocked

VALID SQL RATE (VSR)

98.5%

Syntactically valid AST-verified SQL queries across 100 test prompts

EXECUTION ACCURACY (EX)

92.0%

Exact ground-truth relational result match on multi-table ERP joins

MEDIAN LATENCY (p_{50})

100.9ms

Tail latency
$p_{95} = 119.8$ ms
for end-to-end tool routing & validation

CONTINUOUS INTEGRATION VERIFICATION

100% Pass Rate Across All 6 Individual Team Engineering Slice Suites

✓ All Assertions Passing

PASS	PASS	PASS	PASS	PASS	PASS
`test_moustafa_slice`	`test_adel_slice`	`test_abdallah_slice`	`test_retaj_slice`	`test_renad_slice`	`test_roaa_slice`
LangGraph, AST & HITL	RLS, Auth & Order State	GenUI Cards & RTL UI	LightGBM, SHAP & ROP	Log-RFM & Benchmark	Policy RAG & Guardrails

SLIDE 15 • STRATEGIC PRODUCT EVOLUTION

Beyond Graduation: Scaling TalkShop Across MENA

3-Horizon Enterprise Expansion Plan

1 Q3–Q4 2026: ETA Tax + Voice AI — 2 2027: Omnichannel Social Copilot — 3 2028+: Cross-Tenant Federated ML

Horizon 01 • Near-Term (Q3–Q4 2026)

ETA e-Invoicing & Egyptian Voice Copilot

Direct cryptographic integration with the **Egyptian Tax Authority (ETA)** e-Receipt/e-Invoicing portal alongside hands-free dialectal voice commands.

- Automated ETA Tax UUID signing on order completion
- Fine-tuned Whisper-Large-v3 for Egyptian Arabic speech-to-intent

Phase I — الربط مع الفاتورة الإلكترونية والأوامر الصوتية

Horizon 02 • Medium-Term (2027)

Unified Social Commerce Copilot

Autonomous omnichannel agent converting unstructured shopper chats on **WhatsApp Business**, **Instagram DMs**, and **Messenger** into verified ERP orders.

- Real-time inventory reservation & Paymob payment link generation in DMs
- Multimodal visual product search from customer screenshots

Phase II — تحويل محادثات واتساب وإنستجرام لطلبات آلية

Horizon 03 • Long-Term (2028+)

Cross-Tenant Federated Learning

Privacy-preserving collaborative demand forecasting across thousands of stores without ever sharing or centralizing raw merchant sales records.

- Encrypted gradient aggregation solving cold-start for new stores
- Differential privacy guarantees preserving strict tenant isolation

Phase III — التعلم الفيدرالي المشترك مع حفظ الخصوصية

SLIDE 16 • CONCLUSION & LIVE DEFENSE STAGE

Conclusion & Live TalkShop System Demonstration

Class of 2026 • Data Science & AI Department

KEY CONTRIBUTIONS & MILESTONES

What We Achieved in TalkShop

First Production-Grade Agentic ERP for Egypt

Native Egyptian dialect understanding integrated with Paymob, Bosta, and Consumer Protection Law 181/2018.

Elimination of Manual SQL & Margin Risk

5-layer AST-sandboxed Text-to-SQL (100% security block rate) + Human-in-the-Loop gross margin protection.

High Academic Rigor Across AI Disciplines

Unifies LangGraph State Machines, LightGBM Quantile Forecasting (14.2% WAPE), Log-RFM K-Means (0.5828 Silhouette), and 4-bit GGUF Edge AI.

6 Team Slices • 100% Unittest Verified

Production Ready

● LIVE SYSTEM READY FOR INSPECTION

"Now, we invite the esteemed evaluation committee to experience the live platform in action."

ندعو السادة أعضاء لجنة التحكيم الموقرين الآن لتجربة المنصة بشكل حي ومباشر، ومناقشة الكود المصدري والنماذج الإحصائية.

Frontend Command Center & Storefront
`http://localhost:5173`
React 18 + GenUI

Backend API & Interactive Swagger Docs
`http://localhost:8001/docs`
FastAPI + LangGraph

Thank You • شكراً لحسن استماعكم

Q&A Session Open

Moustafa Maher • Adel Ahmed • Abdallah Sameh • Retaj Hakim • Renad Hakim • Roaa Ashraf